CSCI3150 Introduction to Operating Systems

Lecture 18: Final Review

Hong Xu

https://github.com/henryhxu/CSCI3150

Coming to the end…

- Final exam
- Course review

Final exam

- Closed-book. One A4-sized aid sheet allowed, single-sided
- One calculator allowed
- Answer all 6 questions, each with multiple parts
- Exam questions similar to past papers
- Cover all lectures and tutorials, but not assignments
- Focus on the parts after the midterm

Final exam (cont.)

- Q1: Single-choice and short questions
- Q2: CPU scheduling
- Q3: Segmentation and paging
- Q4: Swapping
- Q5: File systems
- Q6: Misc

Some tips

- Go back to the basics
- Understand the why, not just the how
- Be careful with the details, get them right
- Relax, take it easy (don’t emo!). Just another exam (that you won’t remember at all in ~~2 years~~ 2 months!)

Course Review

Three main parts

- Concurrency
  - Lec 3–9
- Memory management
  - Lec 10–12
- Persistence
  - Lec 13–15
- Misc.
  - Virtualization; networking

Concurrency

- Abstraction: Processes, threads
- Locks, mutex
- Condition variables
- Semaphores

Memory

- Memory management
  - Abstraction: Address space
  - Address translation, base-and-bounds, segmentation
  - Paging, page tables, TLB, cache replacement

Persistence

- File systems
  - I/O devices, generic API
  - Simple FS: inode, data blocks, super blocks
  - LFS: out-of-place update, imap, checkpoint region

That’s it! Thank you all!